Self Supervised Learning: MoCo & SimSiam

Deep Learning

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Da un secolo, oltre.

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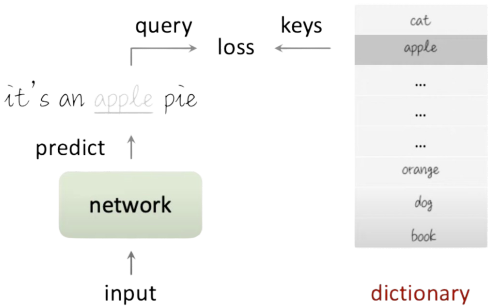
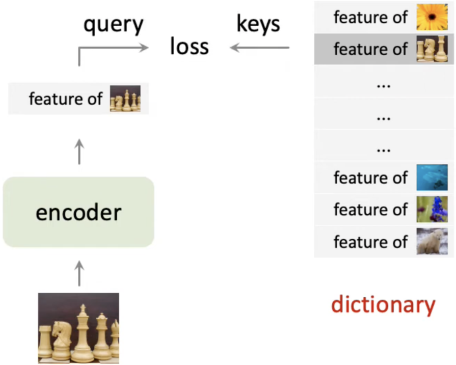
Unsupervised Learning: NLP vs. CV

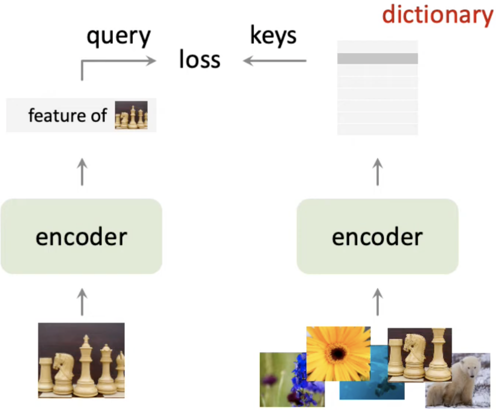
1 MoCo





Contrastive Learning as Dictionary Lookup

1 MoCo



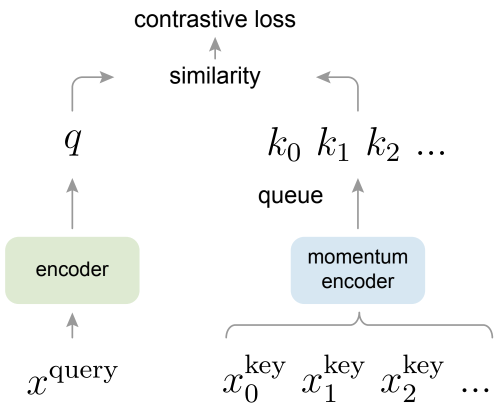
$$\mathcal{L} = -\log \frac{\exp\left(\boldsymbol{q} \cdot \boldsymbol{k}^{+}/\tau\right)}{\sum_{i=0}^{K} \exp\left(\boldsymbol{q} \cdot \boldsymbol{k}_i/\tau\right)}$$

Momentum Contrast

1 MoCo

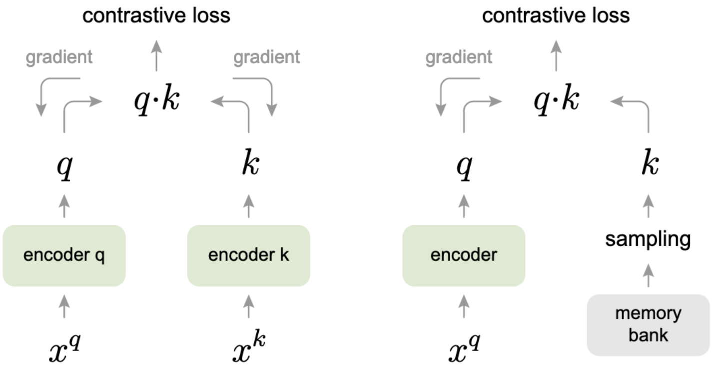
- Large and Consistent dictionary
- Dictionary as a queue of keys features
- Momentum updated key encoder:

$$\theta_k \leftarrow m\theta_k + (1-m)\theta_q$$



Alternative Approaches

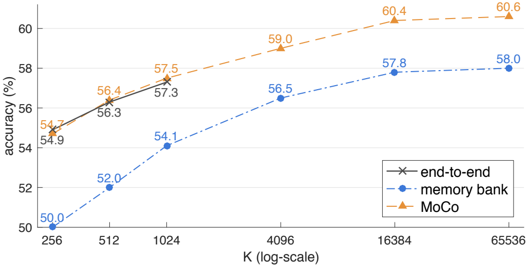
1 MoCo



(a) end-to-end (b) memory bank

Contrastive loss mechanisms

1 MoCo



mechanism	batch	memory / GPU	time / 200-ep.
MoCo	256	**5.0G**	**53 hrs**
end-to-end	256	7.4G	65 hrs
end-to-end	4096	93.0G†	n/a

Table 3. **Memory and time cost** in 8 V100 16G GPUs, implemented in PyTorch. †: based on our estimation.

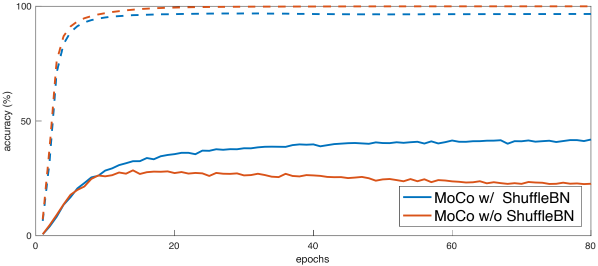
Momentum

momentum m	0	0.9	0.99	0.999	0.9999
accuracy (%)	*fail*	55.2	57.8	59.0	58.9

- The experimental results show that a slowly progressing key encoder is beneficial.

Shuffling Batch Normalization

1 MoCo



- Without shuffling BN, the sub-batch statistics can serve as a signature to tell which sub-batch the positive key is in.

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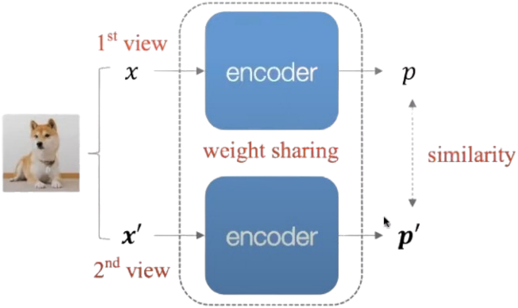
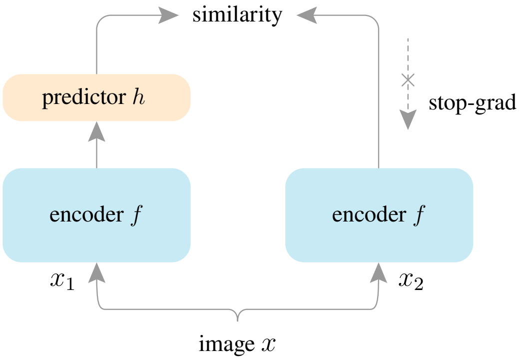
Introduction: Siamese Network



- **Goal**: transfer features to another task
- **Pre-trained features**: should be robust to nuisance factors – Invariance
- **Problem**: Trival/Collapsing solution to a constant

Method

2 SimSiam



- The encoder f consists of a backbone (e.g., ResNet) and a projection MLP head.

$$\mathbf{z}_2 = f(x_2)$$

- The predictor h is an MLP head.

$$\mathbf{p}_1 = h(f(x_1))$$

Loss Function

Minimize the negative cosine similarity between $\mathbf{p}_1$ and $\mathbf{z}_2$:

$$\mathcal{D}(\mathbf{p}_1, \mathbf{z}_2) = -\frac{\mathbf{p}_1}{\|\mathbf{p}_1\|_2} \cdot \frac{\mathbf{z}_2}{\|\mathbf{z}_2\|_2}$$

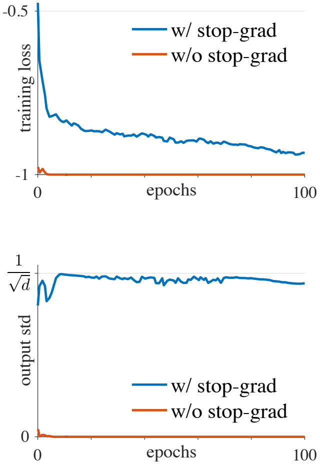
Define a symmetrized loss with a stop-gradient (stopgrad) operation as:

$$\mathcal{L} = \frac{1}{2}\mathcal{D}\big(\mathbf{p}_1, \text{stopgrad}(\mathbf{z}_2)\big) + \frac{1}{2}\mathcal{D}\big(\mathbf{p}_2, \text{stopgrad}(\mathbf{z}_1)\big)$$

Another proposed loss is the Cross-Entropy similarity, obtained by replacing D with

$$D(\mathbf{p}_1, \mathbf{z}_2) = -\text{softmax}(\mathbf{z}_2) \cdot \log \text{softmax}(\mathbf{p}_1)$$

Stop-gradient



Degeneration Without `stopgradient`**:**

- Optimizer converges to loss -1 (degenerate solution).
- Collapse detected by `std`: $\frac{z}{\|z\|_2} \to 0$ (constant vector).
- Isotropic Gaussian baseline: `std` $= \sqrt{\frac{1}{d}}$.

Hypothesis: problem formulation

Consider a loss function of the following form:

$$\mathcal{L}(\theta, \eta) = \mathbb{E}_{x,\mathcal{T}}\left[\|F_\theta(\mathcal{T}(x)) - \eta_x\|_2^2\right]$$

With this formulation, consider solving:

$$\min_{\theta,\eta} \mathcal{L}(\theta, \eta)$$

The problem can be solved by an alternating algorithm

$$\theta_t \leftarrow \arg\min_\theta L(\theta, \eta^{t-1})$$

$$\eta_t \leftarrow \arg\min_\eta L(\theta^t, \eta)$$

Hypothesis: solving

Solving for θ: Can use SGD to solve the sub-problem, the stop-gradient operation is a consequence (η^{t-1} is a constant).
Solving for η: Minimize $\mathcal{L}(\theta, \eta)$ for each image x, approximate by sampling the augmentation only once, denoted as $\mathcal{T}'$:

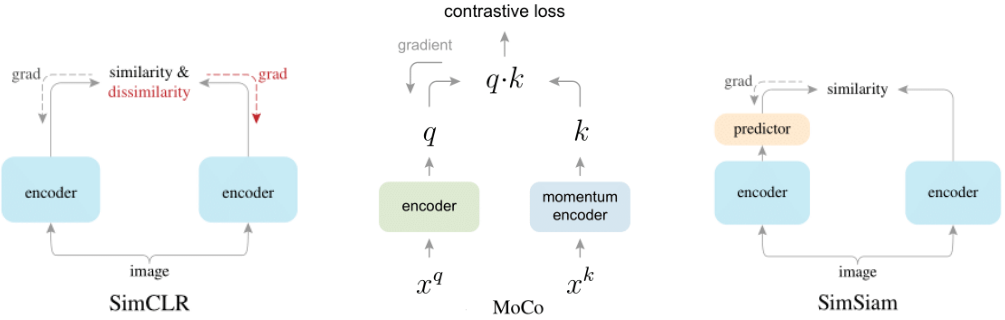
$$\eta_x^t \leftarrow \mathbb{E}_T \left[F_{\theta_t}(\mathcal{T}(x)) \right] \quad \Rightarrow \quad \eta_x^t \leftarrow F_{\theta_t}(\mathcal{T}'(x))$$

One-step alternation:

$$\theta^{t+1} \leftarrow \arg\min_{\theta} \mathbb{E}_{x,\mathcal{T}} \left[\left\| F_\theta(\mathcal{T}(x)) - F_{\theta_t}(\mathcal{T}'(x)) \right\|_2^2 \right]$$

Previous Work

2 SimSiam



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Datasets

We pretrained our models on the MiniImageNet Dataset, and we evaluated our models with the Linear Evaluation protocol for the task of classification on MiniImageNet and CIFAR10.

- **MiniImageNet**: 100 classes, 60000 samples
- **CIFAR10**: 10 classes, 60000 samples

Pretrain settings

Simsiam pretraining:

- Learning Rate: $0.05\times$ batch_size $/256$
- Cosine annealing learning rate scheduler on the backbone and the projector
- Batch Size: 96

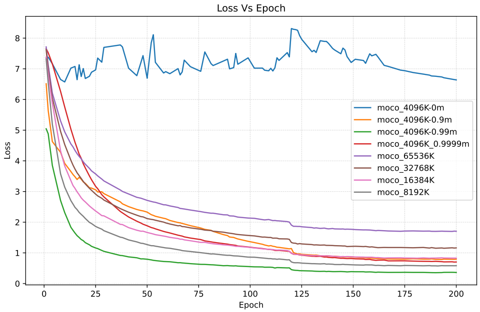
MoCo pretraining:

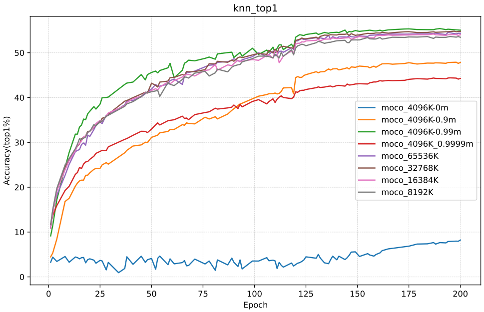
- Learning Rate: 0.03
- MultiStepLR scheduler with $\gamma = 0.1$ and milestones set at 120 and 160 epochs
- Batch Size: 128

Both models were trained for a total of 200 epochs.

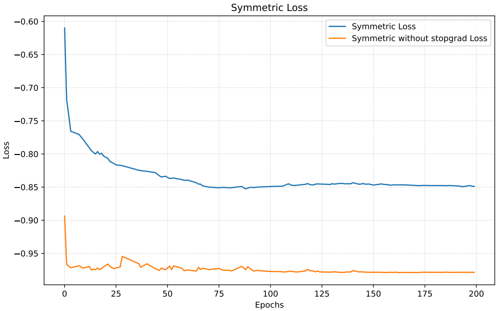
Results: MoCo pretraining

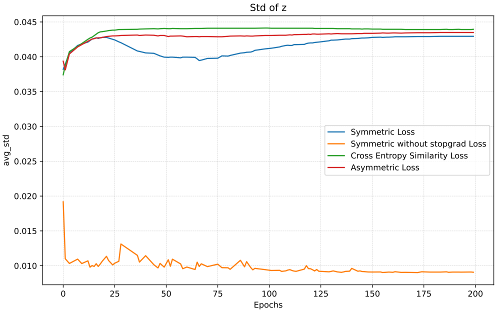
3 Experiments





Results: SimSiam pretraining





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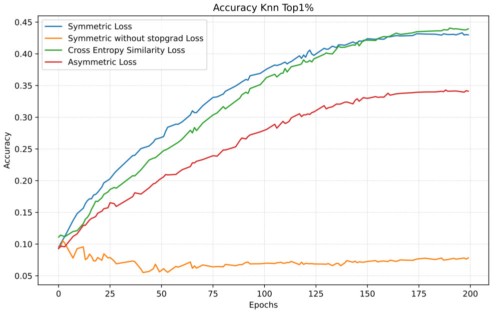
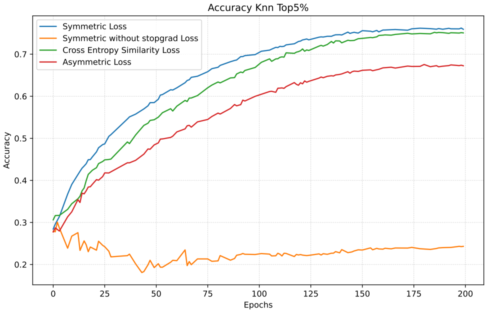
Results: SimSiam pretraining

3 Experiments





Linear evaluation settings

SimSiam Linear evaluation:

- Learning Rate: 0.3 for MiniImageNet, 0.025 for CIFAR10
- Cosine annealing learning rate scheduler

MoCo Linear evaluation:

- Learning Rate: 0.5 for MiniImageNet, 0.05 for CIFAR10
- MultiStepLR scheduler with $\gamma = 0.1$ and milestones set at 60 and 80 epochs

Both models were trained for a total of 100 epochs with batch size set to 128

MoCo Linear evaluation

Accuracy (%)	**Momentum (m)**				
	$m = 0$	$m = 0.9$	$m = 0.99$	$m = 0.999$	$m = 0.9999$
Top1 (MiniImageNet)	fail	53.4%	59.8%	58.5%	50.4%
Top1 (CIFAR10)	fail	49.7%	49.7%	48.5%	46.2%

MoCo Linear evaluation

Accuracy (%)	Dictionary Size (K)				
	$K = 4096$	$K = 8192$	$K = 16384$	$K = 32768$	$K = 65536$
Top1 (MiniImageNet)	58.5%	57.8%	57.6%	58.4%	58.0%
Top1 (CIFAR10)	48.5%	46.8%	48.4%	48.6%	47.9%

SimSiam Linear evaluation

MiniImageNet	Top1 Accuracy		Top5 Accuracy	
	100	200	100	200
Symmetric Loss	52.7%	**57.5**%	80.1%	**83.6**%
Cross Entropy Similarity Loss	50.4%	57.0%	78.1%	83.1%
Asymmetric Loss	45.5%	49.8%	73.5%	78.1%
Symmetric Loss w/o stopgrad	7.7%	6.6%	24.4%	22.2%

SimSiam Linear evaluation

CIFAR10	Top1 Accuracy		Top5 Accuracy	
	100	200	100	200
Symmetric Loss	49.3%	**50.6%**	92.2%	**92.6%**
Cross Entropy Similarity Loss	48.3%	50.5%	92.1%	92.5%
Asymmetric Loss	45.2%	46.9%	90.2%	91.2%
Symmetric Loss w/o stopgrad	21.3%	18.8%	71.3%	68.3%

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Final Result: Moco vs SimSiam

Dataset	MiniImageNet		CIFAR10	
	Top1 Accuracy	Top5 Accuracy	Top1 Accuracy	Top5 Accuracy
MoCo	**59.8%**	**84.3%**	50.0%	92.0%
SimSiam	57.5%	83.6%	**50.6%**	**92.6%**

Links to all conducted experiments:

- MoCo
- SimSiam